Fiscal Year:	FY 2007	**Task Last Updated:**	FY 10/04/2007
PI Name:	Cavanagh, Peter R. Ph.D., D.Sc.		
Project Title:	Foot Reaction Forces During Simulated ISS Exercise Countermeasures		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Bone Loss Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	98195-6500	**Congressional District:**	7
Comments:	PI moved from Cleveland Clinic to University of Washington in June 2008 (8/08)		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 4/30/2008 per K. Major/NSBRI (3/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Perusek, Gail (NASA GRC)		
Grant/Contract No.:	NCC 9-58-BL00402		
Performance Goal No.:			
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	Background: Magnitude of loads (F) and rate of change of loads (dF/dt) are widely believed to be important criteria for efficacy of countermeasures against loss of bone mass. However, it is known that astronauts on the ISS lost lower-extremity bone mass despite exercising. As the current exercise countermeasure devices are used and upgraded during their anticipated 10-year life span, there is a need to establish their loading characteristics in a realistic ground-based simulation so that exercise prescriptions and the design of components such as the harness and subject load device (SLD) can be optimized. In particular, the compliant nature of the interface between the device and its attachment to the ISS has an important influence on F and dF/dt and has not yet been adequately included in previous simulations. The original aims of this project were: (1) To improve the biomechanical validity of the human supine suspension		

Task Description:	approach to microgravity simulation and to develop hardware and software that will characterize the force magnitude (F), rate of change of force (dF/dt), and vibratory and transmission characteristics of load transferred to the human body. (2) To fully characterize lower-extremity loading during high-fidelity simulated 0g exercise on various exercise devices throughout the range of SLD and harness settings. And (3) Based on this new information regarding mechanical "dose", to propose a regimen of exercise on the available devices that will be effective in maintaining bone and muscle mass. To achieve the specific aims of the project, we are conducting modeling, analysis, and human subject experiments to determine the optimal biomechanical simulation of exercise in microgravity. The optimal simulation will take into account the mechanical and inertial conditions of the installation of exercise devices on the ISS. The findings will be generalizable to future countermeasure devices and future space vehicles. Prior Progress: In the first two years of this project, we have completed design and development of an enhanced Zero Gravity Locomotion Simulator (eZLS) at NASA Glenn Research Center to provide a more flight-like simulation of exercise. The project the team at CCF, NASA GRC, and ZIN Technologies completed hardware design, fabrication, qualification, and installation of the device which is now fully operational. Finite element (FE) modeling for all major system components was also performed, together with a system dynamics model in Simulink. The later model was designed to show the influence of vibration developed during exercise on vehicle dynamics. Considerable effort was spent on the development of the human interface aspects of the design. The mechanical design and FEM effort was paralleled with a dynamic modeling effort in MATLAB Simulink, to characterize system dynamic behavior. The team also procured hardware for a servo-motor Subject Load Device (SLD). Other subject loading options have also been pursued including the development of a pneumatic SLD (pSLD). Progress in Year 3: Progress has been achieved on three major objectives during the last year: 1. Subject Load Device: The new subject load device, the pSLD, has now been fully tested and integrated into the ZLS at the Cleveland Clinic. We have established the limits of variability for the device and it has now been moved into routine operation for ongoing experiments. 2. Subject Harness: Considerable progress has been made with the development of a new harness to connect the subject to the subject load device. Through our interactions with backpack manufacturers, and notably Osprey in Cortez, Colorado, a new harness has been developed which will become a high TRL deliverable as part of this project. An evaluation of the harness by two ISS crew members was also completed. 3. Daily Load Stimulus (DLS) Algorithms: The next generation DLS algorithm has been developed, including new theories on saturation and recovery of the osteogenic potential of bone with distributed mechanical loading and the effects of standing. Extensive MATLAB code has been written and is in its final stages of completion, allowing users to analyze complete sets of in-shoe force data. The new algorithm is currently being tested as a method of prescribing an exercise "dose" to subjects in an ongoing bedrest study. 4. Spin-off bedrest exercise device: Our collaborators at Glenn Research Center have leveraged their experience gained on the development of the eZLS into a new device, the standalone ZLS (sZLS) which is about to be delivered to the NASA bedrest facility at the University of Texas Medical Branch in Galveston, TX. Plans for Year 4: In the fourth year of the project, eZLS will be used to examine the role of interface compliance in the foot reaction forces during running. ZLS will be used for a series of human subject experiments to define the range of musculoskeletal loading in cycling, treadmill running, and lunar relevant exercise devices. Through a comparison of data collected from the simulator and previously measured daily loads in 1g, we will provide clear guidelines for the prescription of in-flight exercise that will protect against loss of bone and muscle mass based on a directly measured mechanical "dose." Also in the upcoming year, the modified DLS algorithm will be simulated, tested, and implemented. Further, the harness design will undergo final modifications and requirements for flight readiness will be identified and implemented. Significant progress has been achieved on all three specific aims during the last year. Specific aim three is almost complete and a paper on this topic is an advanced stage of preparation. Activity to address the human subject aspects of specific aims one and two are now in progress and we expect early completion of the specific aim one activities in the second quarter 2007. Experimental work on specific aim number three is currently in progress.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Knowledge gained from this research will contribute to a better understanding of the importance of exercise for the development and maintenance of bone strength among humans living in Earth gravitational fields. Knowledge in this area is crucial to the treatment of bone disease for which exercise may or may not be an effective intervention.
Task Progress:	1. Mechanical Design and Fabrication. Construction and system integration activities for the eZLS are now complete. This facility is now in continuous use at NASA GRC. 2. SLD Servo-Motor System. The project team continued development of the linear motor subject load device (LM-SLD). This SLD has been implemented with sufficient control technology. The LM-SLD is currently being used at NASA GRC for data collection. 3. pSLD. NASA GRC and CCF collaborated to create a passive pneumatic subject load device (pSLD) as a backup to the LM-SLD. This device uses a single pneumatic cylinder to provide a nearly constant spring force on the cables attached to the subject harness. This system has been successfully installed at the Cleveland Clinic and is being used for human subject testing for Specific Aim 2. 4. Harness Design. The SLD and harness are key components in the optimization of subject interface loading characteristics. Discomfort while wearing the harness directly impacts the efficacy of the exercise protocols. In previous years, we explored design changes in the present harness to improve subject comfort, consulted with NASA JSC, and held a meeting with leaders of the backpack industry. In Year 3, fabrication of multiple harnesses for feasibility and comfort testing was completed with partial support from the John Glenn Biomedical Engineering Consortium (JGBEC). An evaluation of the new harness by two ISS crewmembers was also conducted. The harness will be used during human subject testing for this project and is now moving towards a flight readiness review.

	5. System Dynamic Modeling. A dynamic model of the treadmill rack and variable compliant interface hardware was integrated with a human foot impact model. Our integrated model results showed that with a vibration isolation system installed the peak reaction force on the foot could drop by up to 14% compared to a grounded system. The model will be used to predict how prescriptions can be adjusted to reach the desired DLS in a reduced gravity environment. 6. Exercise Prescriptions. The project team has modified existing mathematical theories on daily load stimulus (DLS) to develop a comprehensive algorithm, including new theories on saturation and recovery of the osteogenic potential of bone with distributed mechanical loading and the stimulus provided through standing. The revised algorithm examines both the magnitude domain of the peak loads and the time domain (cyclic nature of typical loading). The revised algorithm has been implemented into a toolkit for the processing of test subject ground reaction force data. 7. Stand-alone Zero Gravity Locomotion Simulator (sZLS). Our collaborators at NASA GRC have leveraged their experience gained on the development of the eZLS into a new device, the Stand-alone Zero Gravity Locomotion Simulator (sZLS) which is about to be delivered to the NASA bedrest facility at UTMB in Galveston, TX. This is a direct spinoff of the NSBRI project.
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